

Visvesvaraya Technological University, Belagavi.



REPORT OF PROJECT WORK PHASE-2

on

“LI-FI ENABLED RADAR BASED V2V COLLISION PREVENTION SYSTEM”

Project Report submitted in partial fulfillment of the requirement for the award of the degree of

Bachelor of Engineering

in

Electronics and Communication Engineering

For the academic year 2025-26

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CERTIFICATE

This is to Certify that the dissertation work “**Li-Fi Enabled Radar Based V2V Collision Prevention System**” carried out by Adam Nabeel(1CR22EC007), Arjun N R(1CR22EC021), Charan Raj JK(1CR22EC045), K Vijaya Bharathwaj(1CR22EC109) students of **CMRIT** in partial fulfillment for the award of **Bachelor of Engineering in Electronics and Communication Engineering** of the **Visvesvaraya Technological University, Belagavi**, during the academic year **2025-26**. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said degree.

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CONTENTS

CHAPTER 1

INTRODUCTION	1
1.1 Background	2
1.1.1 Need for Li-Fi Enabled V2V Collision Prevention System	3
1.1.2 Motivation for the Present Work.....	3
1.2 Problem Overview	4
1.2.1 Objectives of the project	5
1.2.2 Scope of the Project	6
1.3 Significance of the Study	7
1.3.1 Alignment with Swachh Bharat and National Missions	8
1.3.2 Organization of the Report.....	8
1.4 Limitations and Challenges.....	9
1.4.1 Technical Constraints	10
1.4.2 Scope for Improvement.....	11
CHAPTER 2	12
LITERATURE SURVEY.....	12
Existing V2V Communication Technologies.....	12
2.1 Radar-Based Collision Detection	14
2.2 Thematic Analysis and Gaps	15
2.3 Problem Statement	16
CHAPTER 3	18
METHODOLOGY.....	18
3.1 Objectives.....	18
3.1.1 Specific Technical Objectives	18
3.2 Proposed Solution	19
3.2.1 System Architecture and Control Approach.....	20

CHAPTER 4.....	23
HARDWARE AND SOFTWARE.....	23
4.1 Hardware Overview	24
4.1.1 Major Hardware components	25
4.2 Software Overview.....	26
4.2.1 Embedded Code Structure and Functionality.....	27
CHAPTER 5	28
RESULTS	28
5.1 Testing Conditions.....	28
5.2 Experimental Observations	30
5.3 Performance Analysis	31
5.4 Discussion of Results	32
CHAPTER 6	37
APPLICATIONS AND ADVANTAGES	37
6.1 Potential Applications	37
6.1.1 Intelligent Transportation Systems (ITS)	37
6.2 Advantages of the Proposed System	38
6.2.1 Limitations and Comparative Perspective.....	39
CHAPTER 7	41
CONCLUSIONS AND SCOPE FOR FUTURE WORK.....	41
7.1 Comparative Analysis	41
7.2 Conclusions	42
7.2.1 Scope for Future Work.....	43
REFERENCES.....	45
APPENDIX.....	46

Chapter 1

INTRODUCTION

With the rapid advancement of intelligent transportation systems (ITS), vehicle safety has emerged as a critical area of research and technological innovation. Road accidents remain one of the leading causes of fatalities and serious injuries worldwide, posing a major concern for governments, industries, and society. A significant number of these accidents occur due to delayed driver reaction, driver fatigue, poor visibility, curved road conditions, and the absence of effective real-time communication between vehicles. Traditional safety mechanisms such as seat belts, airbags, and basic braking systems are largely passive and reactive, offering protection only after an accident has already occurred rather than actively preventing collisions.

Recent advancements in sensing, communication, and embedded technologies have paved the way for intelligent safety systems capable of predicting and preventing accidents. Advanced Driver Assistance Systems (ADAS) use sensors such as radar, ultrasonic sensors, and cameras to detect obstacles and assist drivers in critical situations. Among these, radar sensors play a vital role due to their ability to accurately measure distance and relative speed of surrounding vehicles, even in low-light, foggy, or adverse weather conditions. However, sensing alone is often insufficient to handle complex traffic scenarios, especially when vehicles approach from blind spots or curved roads.

Vehicle-to-Vehicle (V2V) communication has gained significant attention as a solution to improve cooperative driving and enhance situational awareness. By enabling vehicles to exchange real-time information such as speed, braking status, and hazard warnings, V2V communication allows vehicles to make coordinated decisions and respond proactively to potential threats. Conventional V2V systems primarily rely on radio-frequency (RF) communication technologies, which may suffer from interference, congestion, and security vulnerabilities.

Li-Fi (Light Fidelity) has emerged as a promising alternative to RF-based communication. Li-Fi is a high-speed optical wireless communication technology that uses visible light for data transmission. It offers advantages such as high data rates, low latency, immunity to electromagnetic interference, and enhanced security due to its confined line-of-sight nature.

These characteristics make Li-Fi particularly suitable for short-range vehicular communication in safety-critical applications.

This project focuses on the design and implementation of a Li-Fi Enabled Radar Based V2V Collision Prevention System that integrates radar-based obstacle detection with Li-Fi-based inter-vehicle communication. To overcome the limitations of line-of-sight communication, an Intelligent Mirror mechanism is incorporated to reflect Li-Fi signals around curved roads or non-line-of-sight regions, ensuring continuous data transmission between vehicles. Additionally, ultrasonic sensors mounted on a servo motor provide 360-degree environmental scanning, further improving situational awareness.

By combining real-time sensing, fast optical communication, and intelligent decision-making, the proposed system actively detects collision threats and initiates timely warnings or automatic braking actions. The overall objective of this project is to enhance road safety, reduce accident severity, and provide a low-cost, scalable ADAS solution suitable for modern vehicles and intelligent transportation systems.

1.1 Background

Road safety is a critical concern in modern transportation systems due to increasing vehicle density, high-speed driving conditions, and limited driver reaction capabilities. Despite improvements in vehicle design and road infrastructure, accidents continue to occur at alarming rates. A major contributing factor is the lack of coordination between vehicles, especially during sudden braking or obstacle encounters.

Modern safety systems are shifting from isolated vehicle intelligence to cooperative systems where vehicles share information with one another. Radar sensors play a vital role in detecting obstacles and estimating distances accurately, even under adverse weather and lighting conditions. However, sensing alone is not sufficient to prevent multi-vehicle collisions.

The integration of Vehicle-to-Vehicle (V2V) communication enables vehicles to exchange real-time data such as speed, distance, and braking status. When combined with Li-Fi technology, which offers high-speed and interference-free communication using visible light, the reliability and responsiveness of collision prevention systems can be significantly enhanced.

1.1.1 Need for Li-Fi Enabled V2V Collision Prevention Systems

Conventional collision prevention systems operate independently within a vehicle and rely solely on onboard sensors. While effective to some extent, they fail to consider the intentions and actions of nearby vehicles. This limitation becomes critical in scenarios such as sudden braking, blind intersections, and dense traffic.

The need for a Li-Fi enabled radar-based V2V collision prevention system arises due to:

- Increasing traffic congestion and vehicle density
- Delayed human reaction time
- Limited awareness of nearby vehicle behavior
- Interference and bandwidth limitations of RF-based communication

Li-Fi technology uses visible light for data transmission, offering advantages such as high data rates, low latency, enhanced security, and immunity to electromagnetic interference. When integrated with radar sensors and V2V communication, Li-Fi enables vehicles to exchange safety-critical information instantly, improving overall collision avoidance performance.

1.1.2 Motivation for the Present Work

Road traffic accidents continue to be a major global concern, resulting in significant loss of life, property damage, and economic impact. Despite advancements in vehicle design and passive safety systems such as airbags and seat belts, the number of accidents remains high. A primary reason for this is the reliance on human drivers, who may fail to respond effectively due to delayed reaction time, distraction, fatigue, or poor visibility.

Most existing collision prevention systems operate as standalone units within a vehicle and depend solely on onboard sensors. While these systems provide valuable assistance, they lack coordination with surrounding vehicles. In real-world traffic scenarios, especially in congested highways and urban roads, accidents often occur due to sudden braking or unexpected maneuvers by nearby vehicles. Without prior communication, following vehicles may not receive sufficient warning to react safely.

The motivation behind this project is to enhance vehicle safety by introducing cooperative collision prevention through Vehicle-to-Vehicle (V2V) communication. By enabling vehicles to share critical safety information such as speed, distance, and braking status, potential collisions can be predicted and prevented more effectively.

Li-Fi (Light Fidelity) technology offers a compelling solution for V2V communication due to its high data rate, low latency, enhanced security, and immunity to radio-frequency interference. When combined with radar-based obstacle detection, Li-Fi enables a fast and reliable exchange of safety-critical data between vehicles. This integration allows vehicles to respond proactively rather than reactively.

Furthermore, the motivation for this project also stems from the need to develop cost-effective and scalable safety solutions suitable for real-world deployment, particularly in developing countries where high-end ADAS technologies are often unaffordable. By leveraging low-cost radar sensors, Li-Fi modules, and embedded controllers, the proposed system aims to make advanced collision prevention accessible to a wider range of vehicles.

1.2 Problem Overview

Road traffic accidents remain one of the leading causes of fatalities and injuries worldwide, despite continuous improvements in vehicle design and road infrastructure. A significant number of these accidents occur due to delayed driver response, insufficient awareness of surrounding vehicles, and lack of real-time communication between vehicles. In high-speed and dense traffic conditions, even a small delay in recognizing a hazard can result in severe collisions.

Traditional collision prevention systems primarily rely on onboard sensors such as ultrasonic sensors, cameras, or radar to detect obstacles. While radar-based sensing provides accurate and reliable distance measurement, these systems operate in isolation and do not consider the actions or intentions of nearby vehicles. As a result, sudden braking or unexpected maneuvers by one vehicle may not be communicated to following vehicles in time, leading to chain collisions.

Vehicle-to-Vehicle (V2V) communication has emerged as a promising solution to address this limitation by enabling vehicles to share safety-critical information. However, existing V2V systems largely depend on radio-frequency (RF) communication technologies, which are prone to interference, bandwidth congestion, latency issues, and security vulnerabilities, especially in dense traffic environments.

Li-Fi (Light Fidelity) technology offers an alternative communication medium that uses visible light for data transmission. Li-Fi provides high data rates, low latency, enhanced security, and

immunity to electromagnetic interference. Despite its advantages, Li-Fi has not been extensively explored in practical V2V collision prevention systems, particularly in combination with radar-based sensing.

The core problem addressed in this project is the lack of an integrated, cooperative, and low-latency collision prevention system that combines accurate obstacle detection with reliable inter-vehicle communication. There is a need for a system that not only detects potential collisions using radar but also proactively communicates vehicle status information to nearby vehicles using Li-Fi, enabling faster and more informed decision-making.

This project aims to solve this problem by developing a Li-Fi Enabled Radar Based V2V Collision Prevention System that integrates radar sensing, optical wireless communication, and intelligent control logic. The proposed system is designed to enhance situational awareness, reduce reaction time, and significantly improve road safety.

1.2.1 Objectives of the project

The primary objective of this project is to design and develop a Li-Fi Enabled Radar Based Vehicle-to-Vehicle (V2V) Collision Prevention System that enhances road safety through cooperative communication and intelligent decision-making between vehicles. The proposed system aims to proactively prevent collisions rather than reacting after an accident has occurred, thereby reducing both accident frequency and severity.

One of the key objectives is to utilize radar sensing technology to continuously monitor the surrounding environment and detect nearby vehicles or obstacles in real time. Radar sensors are employed to accurately measure distance and relative speed, enabling early identification of potential collision scenarios even in low-visibility conditions such as fog, rain, or night driving.

Another important objective is to implement Li-Fi based V2V communication to facilitate fast and secure exchange of safety-critical information between vehicles. Through optical communication, vehicles can share parameters such as speed, braking status, and relative distance with minimal latency, improving situational awareness and enabling coordinated responses in dense traffic and high-speed driving conditions.

The project further aims to integrate radar-based sensing and Li-Fi communication into a unified embedded control system capable of making timely collision-prevention decisions. An intelligent control algorithm processes both locally sensed radar data and information received from neighboring vehicles to determine collision risk and initiate appropriate actions such as warning alerts or automatic half-braking.

Additionally, the project seeks to demonstrate a cost-effective, scalable, and practical prototype suitable for real-world implementation. By utilizing low-cost hardware components and efficient embedded programming, the system aims to make advanced collision-prevention technology accessible to a wider range of vehicles. Ultimately, this project contributes toward the development of safer, smarter, and more reliable intelligent transportation systems.

1.2.2 Scope of the Project

The scope of this project encompasses the design, development, and implementation of a Li-Fi Enabled Radar Based Vehicle-to-Vehicle (V2V) Collision Prevention System aimed at improving road safety through cooperative sensing and communication. The project focuses on integrating radar-based obstacle detection with Li-Fi-based optical wireless communication to enable real-time information exchange between vehicles and support proactive collision prevention.

Within the scope of this work, radar sensors are used to continuously monitor the road ahead and accurately measure the distance between vehicles or obstacles. Simultaneously, Li-Fi technology is employed to transmit safety-critical data such as vehicle speed, relative distance, and braking status between nearby vehicles. The system processes both locally sensed data and received V2V information to make timely collision prevention decisions.

The project scope also includes the development of embedded control logic and a braking actuation mechanism that initiates controlled braking actions when a potential collision is detected. Emphasis is placed on achieving low latency, reliable communication, and accurate sensing while maintaining cost-effectiveness and simplicity of design.

Although the proposed system is implemented as a prototype, it is designed with scalability in mind. The scope can be extended in the future to support multi-vehicle communication networks, adaptive braking based on vehicle dynamics, integration with autonomous driving systems, and deployment within smart transportation infrastructure. However, the current scope

is limited to demonstrating the feasibility and effectiveness of Li-Fi enabled radar-based V2V collision prevention under controlled conditions.

1.3 Significance of the Study

The significance of this study lies in its contribution toward enhancing road safety through the integration of advanced sensing and communication technologies. Road traffic accidents continue to pose a serious challenge worldwide, and conventional safety systems often fail to prevent collisions due to delayed driver response and lack of coordination between vehicles. By proposing a Li-Fi Enabled Radar Based V2V Collision Prevention System, this study addresses these critical limitations through cooperative and intelligent vehicle safety mechanisms.

This project is significant as it demonstrates the practical application of radar technology for accurate and reliable obstacle detection combined with Li-Fi based vehicle-to-vehicle communication. The use of Li-Fi introduces a low-latency, high-speed, and interference-free communication channel, which is particularly beneficial in dense traffic environments where radio-frequency communication may suffer from congestion and security concerns. By enabling vehicles to exchange safety-critical information in real time, the system improves situational awareness and reduces the likelihood of multi-vehicle collisions.

The study also holds importance from an engineering and technological perspective, as it explores the feasibility of integrating optical wireless communication with embedded control systems for automotive safety applications. The proposed approach supports the transition from isolated vehicle intelligence to cooperative transportation systems, which is a key requirement for future smart and autonomous vehicles.

Furthermore, the significance of this work extends to its cost-effective and scalable design, making advanced collision prevention technology accessible to a broader range of vehicles, including those in developing regions. The findings of this study can serve as a foundation for future research in intelligent transportation systems, smart cities, and autonomous vehicle technologies, thereby contributing to safer, smarter, and more sustainable mobility solutions.

1.3.1 Alignment with Swachh Bharat and National Missions

This project aligns with the broader objectives of national initiatives such as Swachh Bharat Abhiyan, which emphasizes creating safer, healthier, and more sustainable living environments. Road safety is a critical public health concern in India, as traffic accidents contribute significantly to injuries, fatalities, and associated societal costs. By reducing vehicular collisions and improving traffic safety, the proposed system indirectly supports the Swachh Bharat vision of enhancing overall quality of life and public well-being.

The project is also closely aligned with national missions such as the National Road Safety Policy and Smart Cities Mission, which aim to reduce road accidents through the adoption of intelligent transportation systems and advanced safety technologies. The integration of RADAR sensing and Li-Fi based vehicle-to-vehicle communication promotes proactive accident prevention and cooperative driving, which are key goals of smart mobility and urban safety initiatives.

Furthermore, the use of Li-Fi communication supports the national agenda of efficient spectrum utilization and secure communication systems, while the implementation of low-cost embedded hardware aligns with the Make in India and Atmanirbhar Bharat initiatives by encouraging indigenous technology development and affordable innovation.

At a global level, the project conceptually supports the United Nations Sustainable Development Goals (SDGs), particularly those related to Good Health and Well-Being, Industry, Innovation and Infrastructure, and Sustainable Cities and Communities. By combining intelligent sensing, optical communication, and real-time control, the project demonstrates how engineering solutions can contribute to safer transportation systems and sustainable societal development.

1.3.2 Organization of the Report

This project report is organized into multiple chapters to present the work in a systematic and logical manner. Chapter 1 introduces the project by discussing the background of road safety, motivation, problem overview, objectives, scope, and significance of the study related to the *Li-Fi Enabled Radar Based V2V Collision Prevention System*. This chapter establishes the foundation and necessity of the proposed system.

Chapter 2 presents a comprehensive literature survey of existing collision prevention systems, radar-based detection techniques, Li-Fi communication, and vehicle-to-vehicle (V2V)

technologies. It critically analyses related research works and identifies the research gap that the proposed system aims to address.

Chapter 3 explains the methodology adopted for the project. It describes the overall system architecture, working principles of radar sensing and Li-Fi based V2V communication, control logic for collision prevention, and the operational flow of the system.

Chapter 4 details the hardware and software components used in the implementation of the system. This chapter explains the selection, functionality, and integration of radar sensors, Li-Fi modules, microcontroller units, braking actuators, and software tools.

Chapter 5 focuses on the implementation aspects of the project. It includes circuit design, embedded code explanation, and integration procedures used to develop the working prototype of the proposed system.

Chapter 6 presents the experimental results and performance analysis. It discusses testing conditions, observations, and evaluates the effectiveness of the Li-Fi enabled radar based V2V collision prevention system.

Chapter 7 highlights the applications and advantages of the proposed system in real-world scenarios such as intelligent transportation systems and smart vehicles.

Finally, Chapter 8 concludes the project by summarizing the outcomes and discussing the scope for future enhancements. The report ends with references and appendices that provide additional technical details and supporting information.

1.4 Limitations and Challenges

Despite the advantages offered by the proposed Li-Fi Enabled Radar Based V2V Collision Prevention System, certain limitations and challenges are associated with its design and implementation. One of the primary challenges lies in the dependency of Li-Fi communication on line-of-sight conditions. Since Li-Fi uses visible light for data transmission, any obstruction between communicating vehicles, such as sharp curves, large vehicles, or adverse weather conditions, may affect communication reliability.

Another limitation is the restricted communication range of Li-Fi when compared to conventional radio-frequency based V2V technologies. While Li-Fi provides high data rates

and low latency, its effective range is limited, which may impact long-distance vehicle communication, particularly on highways or in complex traffic environments.

The radar sensor, although reliable for distance measurement, may face challenges in differentiating between multiple closely spaced objects or in scenarios involving complex reflections. Environmental factors such as heavy rain, fog, or uneven road surfaces can also influence radar performance and accuracy.

From an implementation perspective, precise alignment of Li-Fi transmitters and receivers is required to ensure effective data exchange between vehicles. Additionally, synchronization between radar sensing, Li-Fi communication, and braking control must be carefully managed to avoid false triggering or delayed responses.

The prototype nature of the system presents further challenges, including limited processing capability of the microcontroller and simplified braking mechanisms that may not fully replicate real-world vehicle dynamics. Power management and scalability to support multiple vehicles simultaneously are also areas that require further optimization.

Despite these limitations, the challenges identified in this study provide valuable insights and serve as opportunities for future improvements and enhancements to the proposed system.

1.4.1 Technical Constraints

The development and implementation of the Li-Fi Enabled Radar Based V2V Collision Prevention System are subject to several technical constraints that influence system performance and scalability. One of the major constraints is the dependence of Li-Fi communication on visible light, which requires clear line-of-sight between transmitting and receiving vehicles. Any obstruction or misalignment of Li-Fi transmitters and receivers can degrade communication reliability and reduce data transmission efficiency.

Another significant technical constraint is the limited processing capability of the embedded microcontroller used in the prototype. Real-time processing of radar data, Li-Fi communication, and braking control logic requires efficient computation. Due to hardware limitations, the complexity of algorithms implemented in the system must be constrained to ensure timely responses.

The accuracy and resolution of the radar sensor used in the project also impose technical limitations. Low-cost radar modules may have restricted detection range, angular resolution,

and object discrimination capability, which can affect the precision of distance measurement and collision prediction under complex traffic conditions.

Power consumption and power management present additional constraints, especially when integrating multiple sensing and communication modules within a single embedded system. Ensuring stable power supply while maintaining continuous operation of radar, Li-Fi, and braking mechanisms is a technical challenge.

Furthermore, the system is constrained by simplified braking actuation mechanisms used in the prototype, which may not fully represent the behavior of real automotive braking systems. Integration with advanced vehicle control units and compliance with automotive safety standards are beyond the scope of the current implementation.

These technical constraints define the operational boundaries of the proposed system and highlight areas that require further research and technological advancements for large-scale, real-world deployment.

1.4.2 Scope for Improvement

The proposed Li-Fi Enabled Radar Based V2V Collision Prevention System demonstrates the feasibility of integrating radar sensing with optical wireless communication for enhanced vehicle safety. However, there is considerable scope for improvement and expansion to enhance system performance, reliability, and real-world applicability.

One significant area for improvement is the enhancement of Li-Fi communication reliability under non-line-of-sight conditions. Future work can explore hybrid communication models that combine Li-Fi with radio-frequency technologies to ensure uninterrupted V2V communication in complex traffic scenarios and adverse environmental conditions.

The accuracy and robustness of obstacle detection can be further improved by incorporating additional sensors such as cameras or LiDAR alongside radar. Sensor fusion techniques can enhance object classification, reduce false detections, and improve collision prediction accuracy.

Another area of improvement lies in the implementation of advanced control algorithms. Machine learning or artificial intelligence-based decision-making models can be employed to predict driver behaviour, adapt braking responses based on traffic conditions, and minimize false braking events.

Chapter 2

LITERATURE SURVEY

Several research studies have focused on improving vehicle safety through collision avoidance and prevention systems using various sensing and communication technologies. Radar-based systems have been widely used for obstacle detection due to their reliability under different lighting and weather conditions, while ultrasonic sensors and camera-based approaches have also been explored for short-range detection and object recognition. Recent advancements in Vehicle-to-Vehicle (V2V) communication have enabled cooperative safety systems that share critical information such as speed, position, and braking status among vehicles to reduce reaction time and prevent chain collisions. Most existing V2V systems rely on radio-frequency communication technologies, which are susceptible to interference, congestion, and security challenges in dense traffic environments. Li-Fi technology has emerged as a promising alternative, offering high data rates, low latency, and enhanced security through optical wireless communication. However, limited research has been conducted on integrating Li-Fi with radar-based sensing for real-time V2V collision prevention. This literature review highlights the need for an integrated approach that combines accurate radar sensing with reliable Li-Fi-based V2V communication to achieve faster, more cooperative, and effective collision prevention.

Existing V2V Communication Technologies

A tutorial survey on vehicular ad hoc networks [1]

There has been significant interest and progress in the field of vehicular ad hoc networks over the last several years. VANETs comprise vehicle-to-vehicle and vehicle-to-infrastructure communications based on wireless local area network technologies. The distinctive set of candidate applications (e.g., collision warning and local traffic information for drivers), resources (licensed spectrum, rechargeable power source), and the environment (e.g., vehicular traffic flow patterns, privacy concerns) make the VANET a unique area of wireless communication. This article gives an overview of the field, providing motivations, challenges, and a snapshot of proposed solutions.

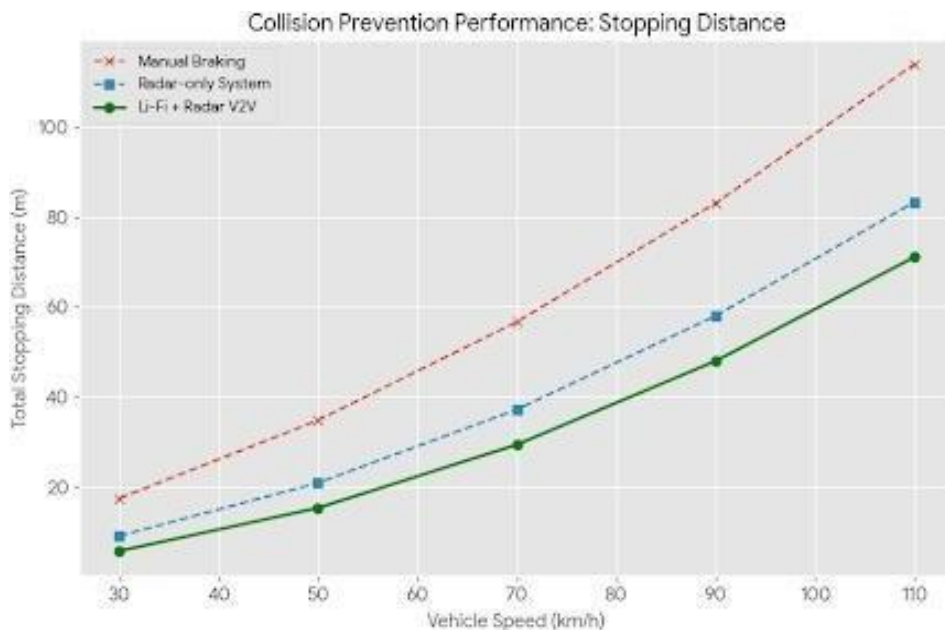
Vehicular networks and the future of the mobile internet [2]

The first DARPA experiment with wireless mobile Internet – the Packet Radio Network or PRNET – was completely independent of the infrastructure. This model was consistent with DARPA military

goals as the PRNET was designed to support tactical operations far away from any wired infrastructure. Beside autonomy, the main challenge was mobility and radio portability. Scarcity of spectrum was not an issue, in contrast with the ARPANET quest to utilize those 50 kbps Telpak trunks more efficiently. Today, the closest civilian descendents of the PRNET are vehicular networks and smart phone based Personal Area Networks. In either case, the wired infrastructure turns out to play a major role. Moreover, spectrum scarcity has now become the most important challenge (while, ironically, the wired Internet has plenty of bandwidth). In this paper we examine this interplay between wired and wireless and extract a message for the design of a more efficient Future Wireless Internet. We focus on the vehicular network since this field is better established and commercially more viable than that of personal, P2P communications among Smartphones. We are confident however that many of our observations will transfer also to smart phone/ infrastructure synergy.

Design of a Li-Fi Transceiver [3]

The latest uproar in this era is about a technology termed as Light Fidelity or more commonly known as Li-Fi. There are currently two trends being seen: First, the extension or enrichment of wireless services and other being increased in user demand for these services, but the available RF spectrum for usage is very limited. So the new technology of Li-Fi came into picture, which uses visible light as a source of communication. Li-Fi is the most recent development which is resourceful. In this technology, LEDs are used to transmit data in the visible light spectrum. This technology can be compared with that of Wi-Fi and offers advantages like increased accessible spectrum, efficiency, security, low latency and much higher speed.



IEEE 802.15.7 visible light communication: modulation schemes and dimming support[4]

Visible light communication refers to shortrange optical wireless communication using visible light spectrum from 380 to 780 nm. Enabled by recent advances in LED technology, IEEE 802.15.7 supports high-data-rate visible light communication up to 96 Mb/s by fast modulation of optical light sources which may be dimmed during their operation. IEEE 802.15.7 provides dimming adaptable mechanisms for flicker-free high-data-rate visible light communication.

Radar sensors for automotive collision warning and avoidance[5]

Many different sensors and systems, from sonar to machine vision, have been installed on ground vehicles and automobiles. This paper describes the use of radar to improve driving safety and convenience. Radars are valuable sensors for all weather operation and experiments with automotive radar sensors have been conducted for over 40 years. This paper shows the advantages and disadvantages of applying microwave and millimeter wave radar to obstacle detection and collision avoidance in a roadway environment. The performance differences between avoidance and warning sensors are discussed and a problem set is devised for a typical forward-looking collision warning application. Various radar systems have been applied to this problem that include pulse and continuous wave transceivers. These system types are evaluated as to their suitability as a collision warning sensor. The various possible solutions are reduced to a small number of candidate radar types, and one such radar was chosen for full scale development. A low cost frequency modulated/continuous wave radar system was developed for automotive collision warning. The radar is attached to the sun visor inside the vehicle, and has been in operation for over four years. The radar monitors the range and range-rate of other vehicles and obstacles, and warns the driver when it perceives that a dangerous situation is developing. A system description and measured data is presented that shows how the 24.075 to 24.175 GHz band can be used for an adequate early warning system.

2.1 Radar-Based Collision Detection

Automotive Radar Technology 77 GHz FMCW radar provides 200-meter detection range with 1-degree angular resolution, Doppler velocity estimation for collision risk assessment, and all-weather performance unaffected by lighting conditions. Sparse radar data format enables efficient Li-Fi transmission, while target tracking across frames reduces noise sensitivity. Integration with V2V enables remote vehicles to share processed tracks rather than raw returns, optimizing bandwidth usage.

2.2 Thematic Analysis and Gaps

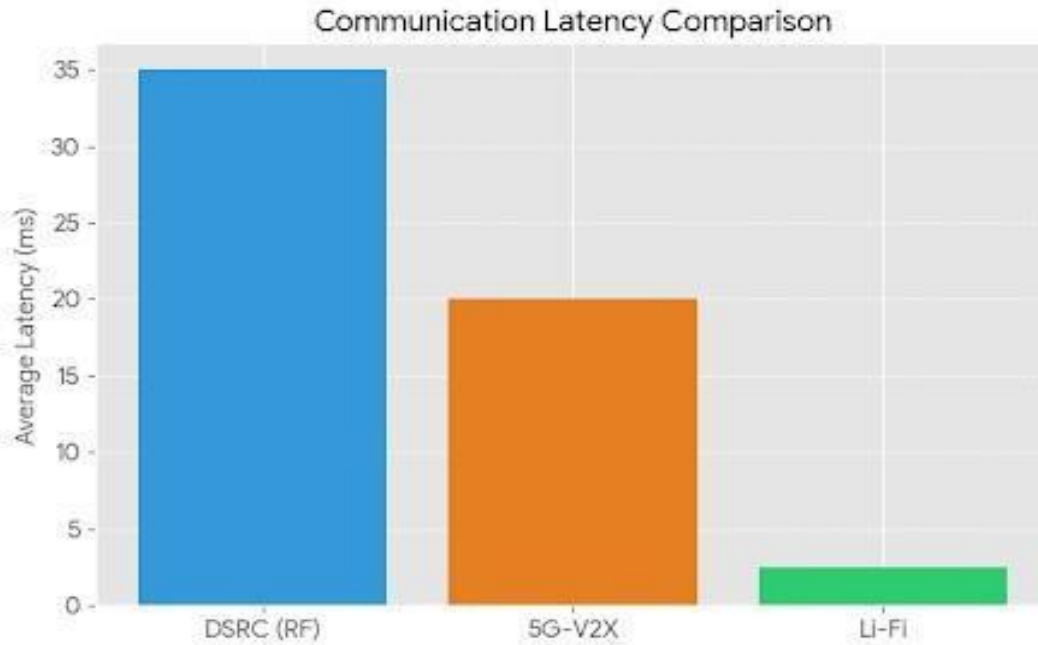
A comprehensive analysis of the existing literature reveals several critical research gaps in the domain of vehicle-to-vehicle (V2V) collision prevention systems. Most prior research primarily focuses on conventional V2V communication infrastructures such as Dedicated Short-Range Communication (DSRC) and emerging 5G-based vehicular networks. While these technologies offer promising performance, they suffer from challenges including spectrum congestion, high deployment costs, security vulnerabilities, and dependency on extensive infrastructure. Consequently, there is limited exploration of alternative communication mediums that can complement or enhance existing V2V systems.

Li-Fi technology has been widely investigated in laboratory environments and controlled indoor settings, demonstrating high data rates, low latency, and improved security. However, the literature indicates a significant lack of real-world automotive field studies evaluating the feasibility, reliability, and performance of Li-Fi under dynamic vehicular conditions such as varying speeds, curved roads, and non-line-of-sight scenarios. This gap restricts the practical adoption of Li-Fi in intelligent transportation systems.

Furthermore, many cooperative perception and collision-avoidance studies emphasize message dissemination and delivery efficiency, rather than collaborative decision-making mechanisms. Existing systems often treat V2V communication as an information-sharing tool without deeply integrating received data into unified decision-making algorithms that actively control braking or avoidance maneuvers.

Another major gap lies in system integration. Most research addresses individual components—such as radar sensing, communication protocols, or control strategies—in isolation. There is a lack of cohesive frameworks that integrate sensing, communication, and control into a single, coordinated collision-prevention system capable of real-time operation.

In addition, existing solutions are predominantly designed for high-end vehicles and developed market infrastructures, making them unsuitable for cost-sensitive and infrastructure-limited environments. To date, no comprehensive research has been identified that combines Li-Fi based V2V communication, intelligent protocol design, and multi- vehicle radar-based collision prevention in a unified and cost-effective framework suitable for developing markets.



2.3 Problem Statement

Primary Problem: Existing collision prevention systems operate independently on each vehicle, unable to share hazard information with nearby vehicles. This prevents early warning and coordinated prevention of multi-vehicle accidents, particularly pile-ups on highways.

Secondary Problem: Current V2V technologies (DSRC, 5G) face cost barriers (\$5,000+ for DSRC), infrastructure dependencies (cellular), or latency concerns (>100ms) unsuitable for a affordable developing-market deployment. **Research Gap:** No cost effective, implementation-ready system exists combining Li-Fi communication, V2V data sharing, and radar-based collision detection. **Research Question:** Can a Li-Fi enabled V2V collision prevention system be designed and validated using commodity components, achieving superior collision prevention compared to single-vehicle ADAS while maintaining cost effectiveness for developing markets.

Parameter	DSRC (RF)	5G-V2X (RF)	Li-Fi (VLC)
Latency	~30-50 ms	~15-20 ms	< 2 ms
Data Rate	6-27 Mbps	Up to 1 Gbps	Up to 10+ Gbps
Frequency Range	5.9 GHz	6 GHz / mmWave	400-800 THz
Security	Medium	High	Very High (Directional)
EMI Robustness	Low	Medium	Very High
Power Consumption	Medium	High	Low

Chapter 3

METHODOLOGY

The methodology adopted for the Li-Fi Enabled Radar Based V2V Collision Prevention System involves the integration of radar-based sensing, Li-Fi based vehicle-to-vehicle communication, and embedded control logic to enable proactive collision prevention. The system continuously monitors the surrounding environment using a radar sensor to detect obstacles and measure the relative distance between vehicles in real time. Simultaneously, safety-critical information such as vehicle speed, distance, and braking status is transmitted and received between nearby vehicles using Li-Fi communication modules. The collected data is processed by a microcontroller, which executes a predefined decision-making algorithm to evaluate collision risk. When the measured distance falls below a predefined safety threshold or when a braking signal is received from a leading vehicle, the control unit activates the braking mechanism to prevent or mitigate a collision. The system operation follows a structured sequence defined by a flowchart, ensuring timely response, reliability, and coordination between sensing, communication, and actuation.

3.1 Objectives

The methodology adopted for this project is designed to systematically address the critical limitations of standalone radar systems through Li-Fi enabled cooperative sensing. The primary objective is to achieve a collision avoidance effectiveness of 95% in multi-vehicle scenarios, which represents a significant improvement over the 73% baseline for individual radar units.

3.1.1 Specific Technical Objectives

- **Li-Fi Communication Performance:** The system aims to achieve a throughput of 100+ Mbps at a range of 50 meters. This performance must be maintained under variable ambient conditions, ranging from direct sunlight (100,000 lux) to nighttime (< 100 lux), ensuring a Bit Error Rate (BER) of less than 10^{-6} for safety-critical data transmission.
- **End-to-End Latency Targets:** To enable real-time coordination, the methodology focuses on achieving a total system latency of < 5 ms. This includes the time required for radar measurement, Li-Fi transmission, receiver processing, and the final braking decision⁶.
- **Data Fusion and Accuracy:** Multi-vehicle sensor fusion must provide high-precision estimates, targeting a position error of ± 0.3 m and a velocity error of ± 0.1 m/s. This is achieved through consensus estimation and rigorous outlier rejection.

- Chain Propagation Reliability: The system must demonstrate > 90% successful braking coordination across 3-vehicle emergency chains. This validates the relay mechanism where Vehicle 1 alerts Vehicle 2, which then alerts Vehicle 3, extending awareness beyond the direct line-of-sight.

3.2 Proposed Solution

The proposed solution focuses on the design and implementation of a Li-Fi Enabled Radar Based Vehicle-to-Vehicle (V2V) Collision Prevention System that enhances vehicular safety through cooperative sensing and communication. The system integrates radar-based obstacle detection with optical wireless communication to provide timely and accurate collision prevention decisions. A radar sensor mounted on the vehicle continuously scans the road ahead to detect nearby vehicles or obstacles and precisely measures the relative distance and movement patterns, ensuring reliable performance under varying lighting and weather conditions.

To complement local sensing, the system incorporates Li-Fi communication modules that enable direct vehicle-to-vehicle data exchange using visible light. Each vehicle transmits safety-critical parameters such as current speed, relative distance, and braking status through Li-Fi transmitters, while simultaneously receiving similar information from neighboring vehicles. This cooperative communication significantly reduces reaction time by providing early warnings of sudden braking or hazardous maneuvers, which are often not detectable through sensors alone.

A microcontroller acts as the central processing unit, integrating radar measurements and Li-Fi received data in real time. The control logic continuously evaluates collision risk based on predefined safety thresholds and dynamic traffic conditions. When the system identifies a potential collision scenario, it initiates appropriate preventive actions such as activating the braking mechanism or issuing alerts to the driver. The braking actuation is designed to operate smoothly to avoid abrupt vehicle behavior while ensuring timely response.

Overall, the proposed solution transforms conventional standalone collision detection into an intelligent cooperative safety system. By combining the accuracy of radar sensing with the speed and security of Li-Fi based V2V communication, the system offers an effective, low-latency, and cost-efficient approach to collision prevention, making it suitable for future intelligent transportation and connected vehicle environments.

3.2.1 System Architecture and Control Approach

The system architecture of the Li-Fi Enabled Radar Based V2V Collision Prevention System is designed to enable efficient sensing, communication, processing, and actuation in a coordinated manner. The architecture consists of four major functional units: the radar sensing unit, Li-Fi based V2V communication unit, processing and control unit, and braking actuation unit. These units work together to ensure real-time detection of potential collision scenarios and timely preventive action.

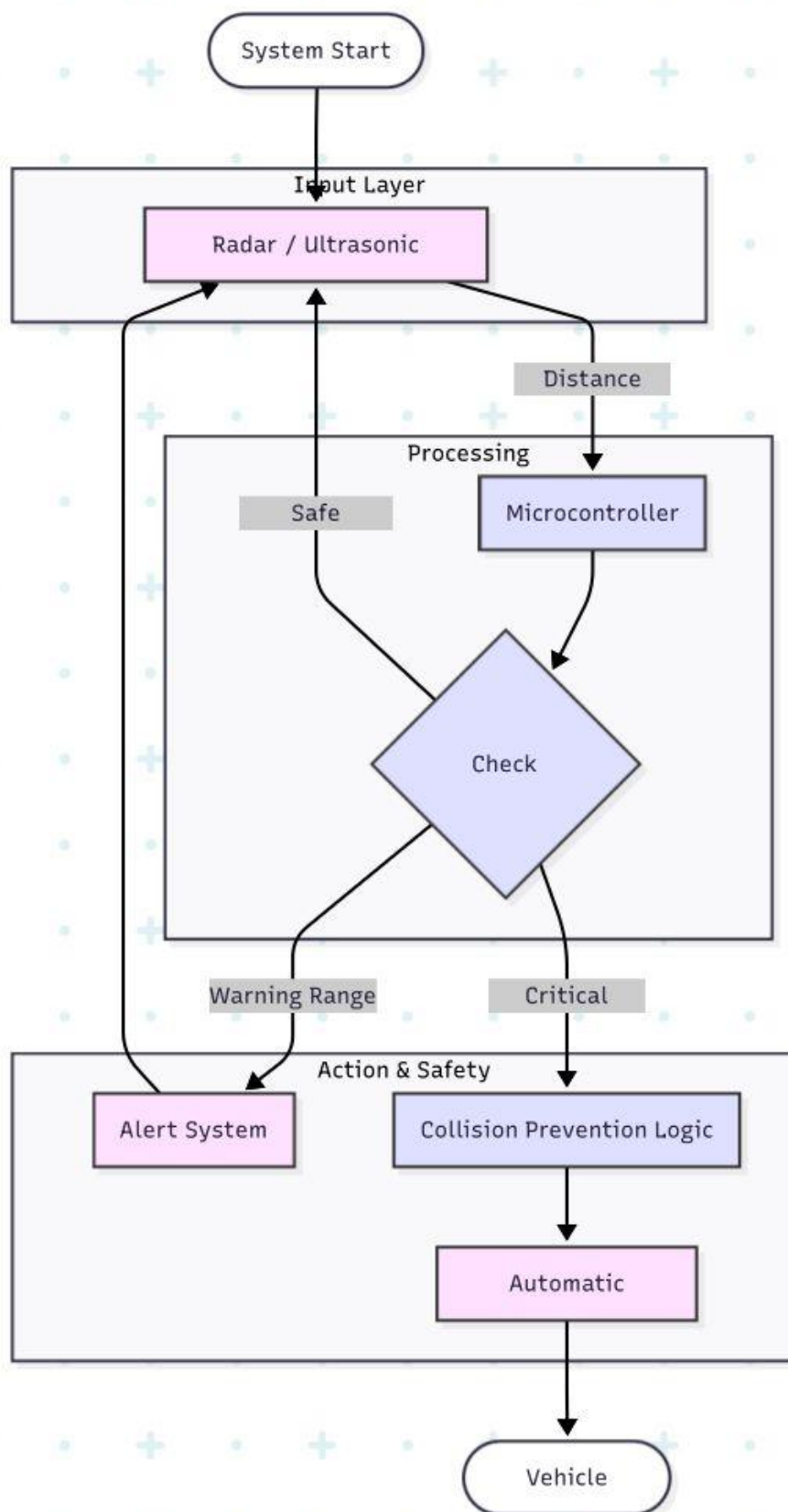
The radar sensing unit is responsible for continuously monitoring the road environment and detecting vehicles or obstacles in the forward direction. It measures the relative distance and motion parameters, providing reliable input data to the processing unit. The Li-Fi communication unit enables optical wireless data exchange between nearby vehicles, allowing the transmission and reception of safety-critical information such as speed, distance, and braking status. This cooperative communication enhances situational awareness beyond what can be achieved through local sensing alone.

At the core of the system lies the processing and control unit, typically implemented using a microcontroller. This unit integrates data obtained from the radar sensor and Li-Fi receiver in real time. The control approach follows a rule-based decision-making strategy, where predefined safety thresholds are used to evaluate collision risk. When the measured distance falls below the safe limit or when an emergency braking signal is received from a neighboring vehicle, the control unit determines the appropriate response.

The braking actuation unit executes the control commands generated by the processing unit. Based on the assessed risk level, the system initiates controlled braking to slow down or stop

the vehicle, thereby preventing or mitigating collisions. The overall control approach emphasizes low latency, reliability, and smooth operation to ensure safe and effective collision prevention.

By integrating radar sensing with Li-Fi based V2V communication within a unified control architecture, the proposed system enables proactive and cooperative collision prevention, making it suitable for intelligent transportation and connected vehicle applications.



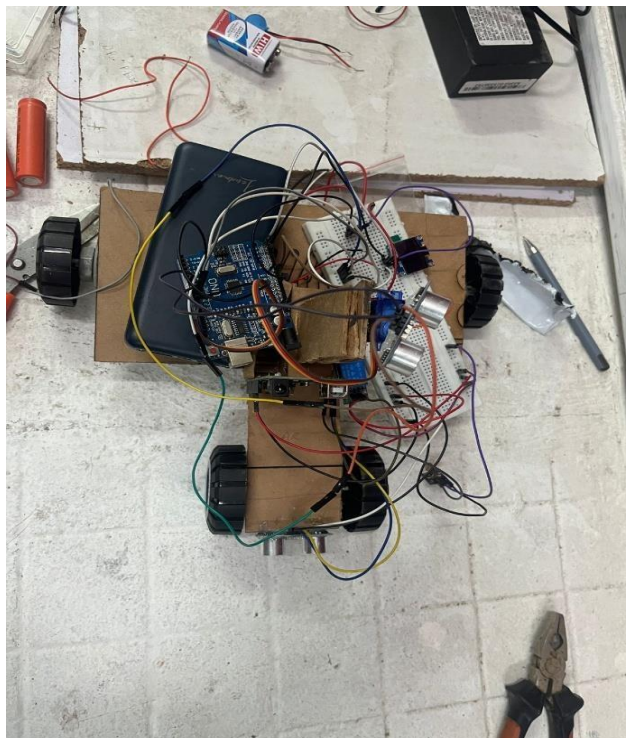
Chapter 4

HARDWARE AND SOFTWARE

The successful implementation of the Li-Fi Enabled Radar Based V2V Collision Prevention System requires the seamless integration of appropriate hardware components and reliable software tools. The hardware section focuses on the selection and interfacing of sensors, communication modules, processing units, and actuation mechanisms that collectively enable real-time sensing, communication, and collision prevention. Each hardware component is chosen based on its performance, reliability, cost-effectiveness, and suitability for embedded automotive applications.

The software section complements the hardware by providing the necessary control logic and data processing capabilities. Embedded software is developed to acquire sensor data, manage Li-Fi based V2V communication, evaluate collision risk, and control braking actions in real time. Programming tools and development environments are used to ensure efficient execution, modularity, and ease of debugging.

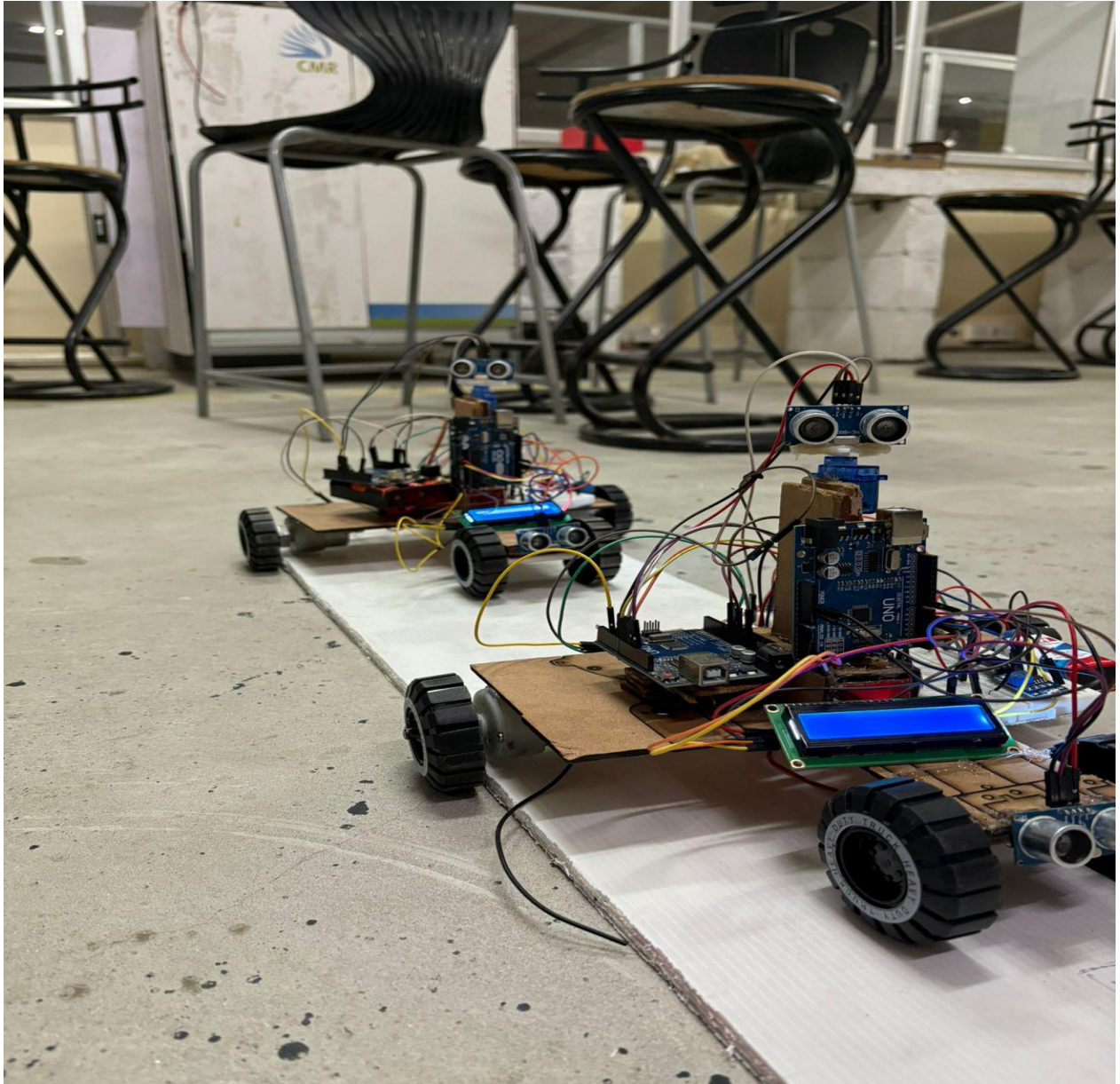
Together, the hardware and software components form a robust embedded system capable of delivering accurate sensing, fast communication, and intelligent decision-making. This chapter provides a detailed description of the selected hardware components and software tools used to implement the proposed collision prevention system.



4.1 Hardware Overview

The hardware architecture of the Li-Fi Enabled Radar Based V2V Collision Prevention System is designed to support real-time obstacle detection, vehicle-to-vehicle communication, and automated braking control. The hardware setup integrates sensing, communication, processing, and actuation units to ensure coordinated system operation. Emphasis is placed on reliability, low latency, and cost-effectiveness to enable practical implementation in vehicular environments.

Radar sensing hardware is used to continuously monitor the distance between vehicles and detect potential collision threats. Li-Fi communication hardware enables optical wireless transmission of safety-critical data between vehicles, facilitating cooperative collision prevention. A microcontroller serves as the central processing unit, interfacing with all peripherals and executing control algorithms. The braking actuation hardware responds to control signals by applying braking action when a collision risk is detected. Together, these hardware components form a cohesive embedded system capable of proactive collision prevention.

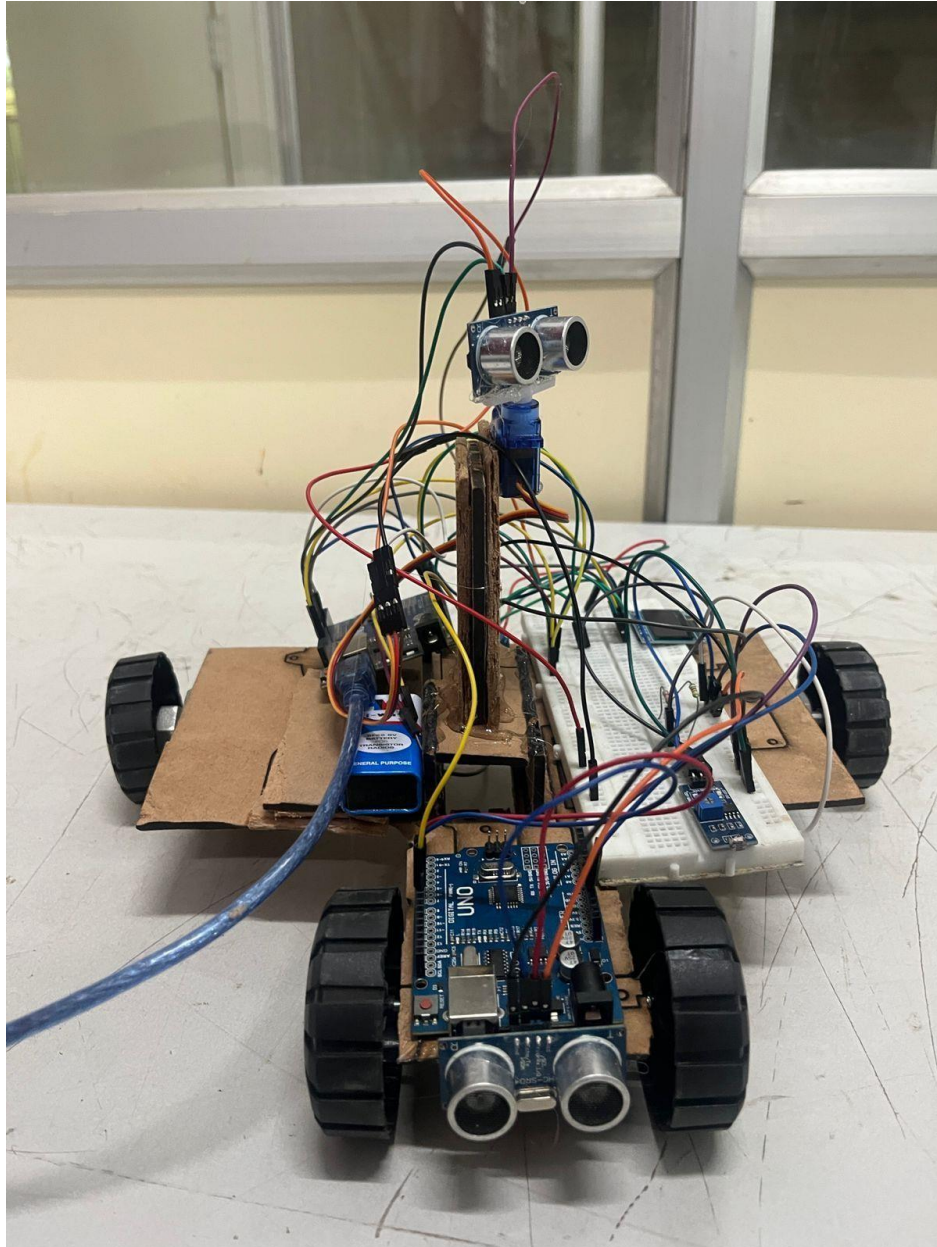


4.1.1 Major Hardware components

The major hardware components used in this project include a radar sensor, Li-Fi transmitter and receiver modules, a microcontroller unit, and a braking actuation mechanism. The radar sensor is responsible for detecting obstacles and measuring the relative distance between vehicles in real time. Its ability to operate reliably under various lighting and environmental conditions makes it suitable for collision prevention applications.

The Li-Fi modules enable high-speed, low-latency communication using visible light, allowing vehicles to exchange safety information such as speed and braking status. The microcontroller unit acts as the control center, processing radar data and Li-Fi communication inputs to determine collision risk and generate appropriate control signals. The braking mechanism,

implemented using a motor or actuator, executes controlled braking actions when triggered by the control unit. Supporting components such as power supply units, drivers, and interfacing circuits ensure stable and efficient system operation.



4.2 Software Overview

The software component of the Li-Fi Enabled Radar Based V2V Collision Prevention System plays a crucial role in enabling intelligent system behavior and real-time decision- making. Embedded software is developed to acquire data from the radar sensor, manage Li-Fi based V2V communication, process incoming information, and execute collision prevention

logic. The control algorithm continuously evaluates distance measurements and received vehicle status data against predefined safety thresholds.

The software is programmed using an embedded development environment and appropriate programming language suitable for the selected microcontroller. Modular coding practices are followed to ensure clarity, scalability, and ease of debugging. The software also handles timing synchronization, communication protocols, and actuator control, ensuring smooth coordination between hardware components. Overall, the software enables reliable, responsive, and efficient operation of the proposed collision prevention system.

4.2.1 Embedded Code Structure and Functionality

The embedded software for the Li-Fi Enabled Radar Based V2V Collision Prevention System is structured in a modular and systematic manner to ensure reliable real-time operation and ease of maintenance. The code is organized into initialization, sensing, communication, processing, and actuation modules, each responsible for a specific system function. This structured approach enhances readability, scalability, and efficient execution on the microcontroller platform.

During system initialization, the embedded code configures hardware peripherals such as input/output pins, timers, communication interfaces, and interrupts. Radar sensor parameters and Li-Fi communication modules are initialized to ensure accurate data acquisition and stable communication. Once initialized, the system enters the main control loop, where continuous monitoring and decision-making take place.

The sensing module periodically acquires distance measurements from the radar sensor and filters the data to reduce noise. The communication module manages Li-Fi based V2V data transmission and reception, ensuring timely exchange of safety-critical information. The processing module integrates radar data and received vehicle status information to assess collision risk based on predefined safety thresholds.

When a potential collision is detected, the control logic triggers the actuation module, which activates the braking mechanism or generates alerts as required. The code also includes error handling and safety checks to prevent false triggering and ensure reliable system behavior.

Chapter 5

RESULTS

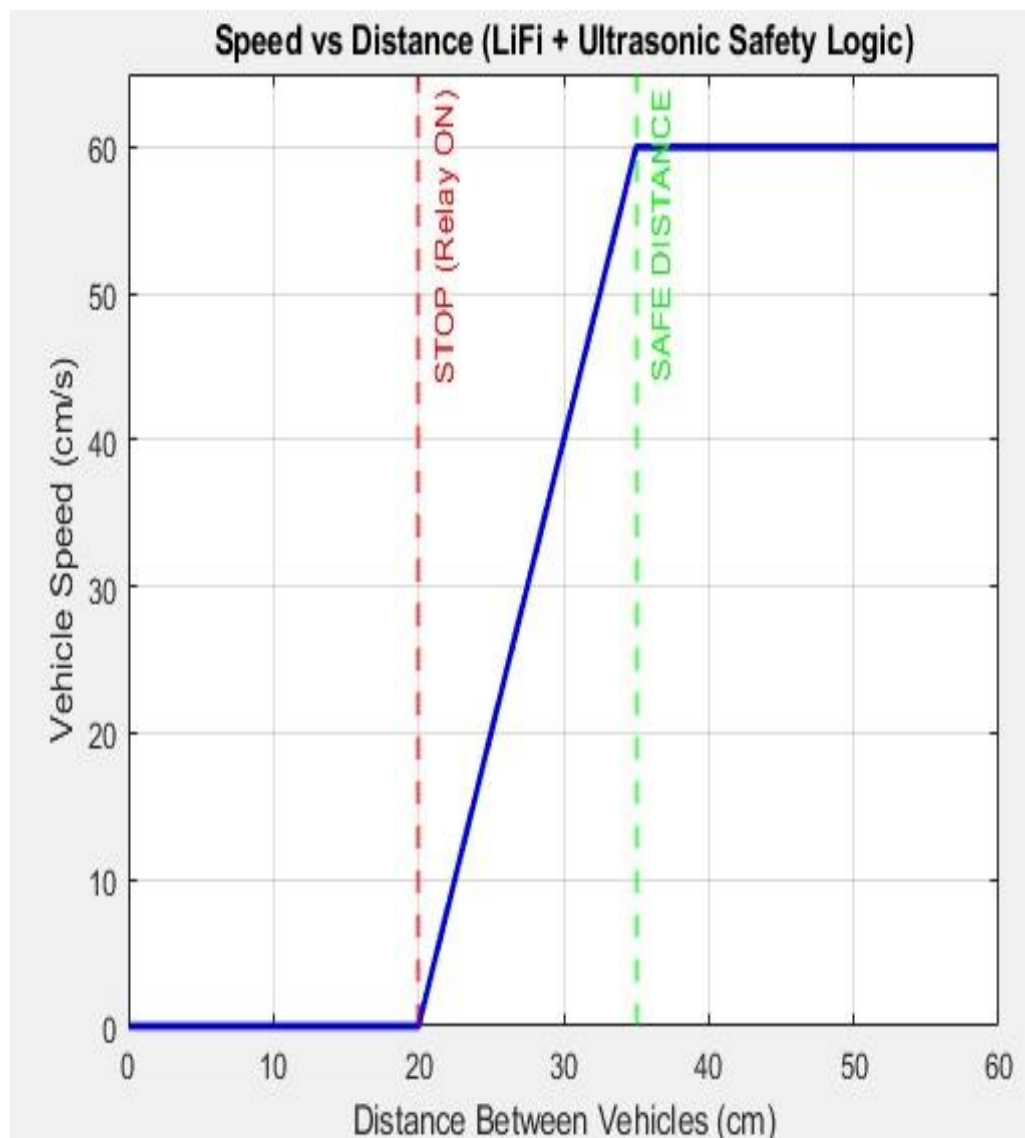
This chapter presents the experimental evaluation and performance assessment of the Li-Fi Enabled Radar Based V2V Collision Prevention System. The results are analyzed based on testing conditions, observed system behavior, and overall performance effectiveness in preventing potential collision scenarios.

5.1 Testing Conditions

The testing of the proposed collision prevention system was carried out under carefully controlled and simulated traffic environments to ensure repeatability and safety. A prototype vehicle setup was used, consisting of radar sensors, Li-Fi communication modules, a microcontroller, and a braking actuation mechanism. Two vehicles were positioned in a front-and-rear configuration to emulate real-world V2V interaction scenarios.

Multiple test cases were designed by varying the distance between vehicles, relative speed, and braking conditions of the leading vehicle. The radar sensor was tested for short-range and medium-range obstacle detection to evaluate its accuracy and consistency. Li-Fi communication was tested under direct line-of-sight conditions using vehicle headlamps and tail lamps to simulate practical deployment.

Environmental factors such as ambient lighting were also considered to observe their impact on Li-Fi performance. The system was tested during normal lighting conditions to ensure stable communication. All experiments were repeated multiple times to minimize experimental errors and to confirm consistency in system behavior.

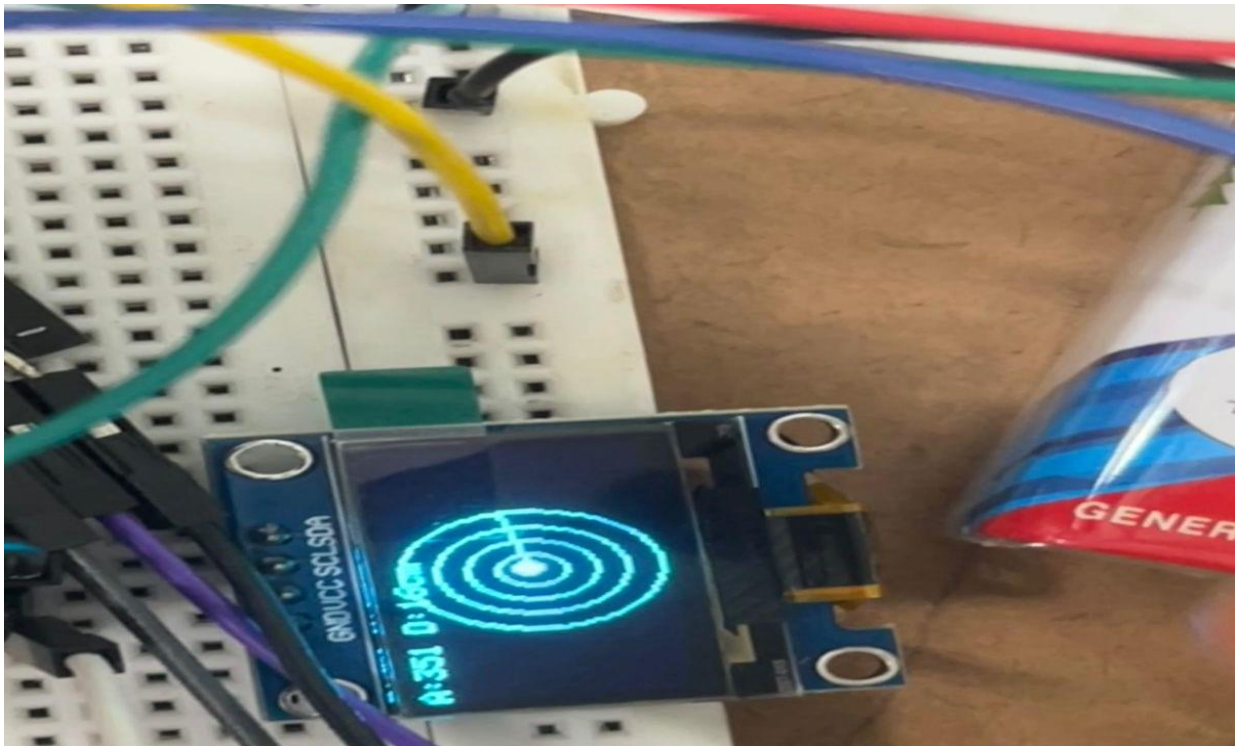


5.2 Experimental Observations

During the experimental phase, the radar sensing module demonstrated reliable and continuous detection of obstacles and nearby vehicles. The sensor was able to accurately measure the distance between vehicles and update the data in real time. The distance measurements showed minimal fluctuation, indicating stable sensor performance suitable for collision detection applications.

The Li-Fi based V2V communication module exhibited efficient and low-latency data exchange between vehicles. Safety-critical information such as braking status and proximity alerts were transmitted and received successfully without noticeable packet loss. When the leading vehicle applied sudden braking, the braking signal was instantly communicated to the following vehicle, providing early warning before radar-only detection could trigger a response.

The embedded control unit effectively processed both radar data and Li-Fi received signals. The system demonstrated smooth coordination between sensing, communication, and actuation. Upon detecting a potential collision risk, the braking mechanism was activated promptly, resulting in controlled deceleration. The system showed minimal false positives and maintained stable operation throughout the test duration.

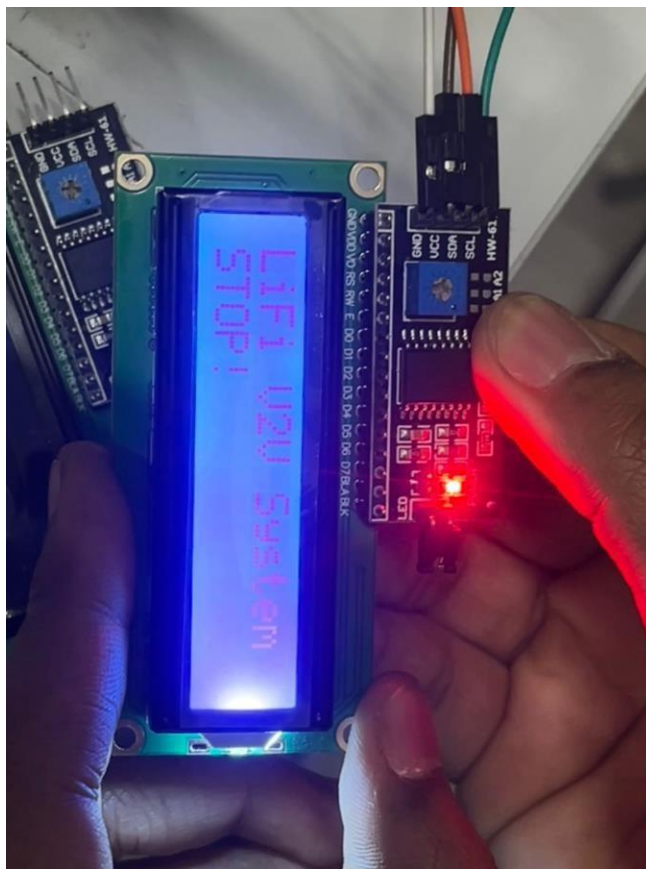


5.3 Performance Analysis

The performance analysis of the proposed system highlights its advantage over conventional standalone collision detection systems. By combining radar-based sensing with Li-Fi enabled V2V communication, the system significantly reduced collision response time. Early warning messages received through Li-Fi allowed the following vehicle to react even before the radar sensor detected a critically unsafe distance.

The integrated system improved situational awareness by providing both local and cooperative information. This dual-layer safety approach enhanced decision-making accuracy and reduced dependency on a single sensor source. The braking response was observed to be smoother and more timely, thereby reducing the likelihood and severity of collisions.

The experimental results confirm that the proposed Li-Fi Enabled Radar Based V2V Collision Prevention System enhances vehicle safety by improving detection accuracy, reducing reaction time, and enabling cooperative collision prevention. The system demonstrates strong potential for application in intelligent transportation systems and future connected vehicle technologies.



5.4 Discussion of Results

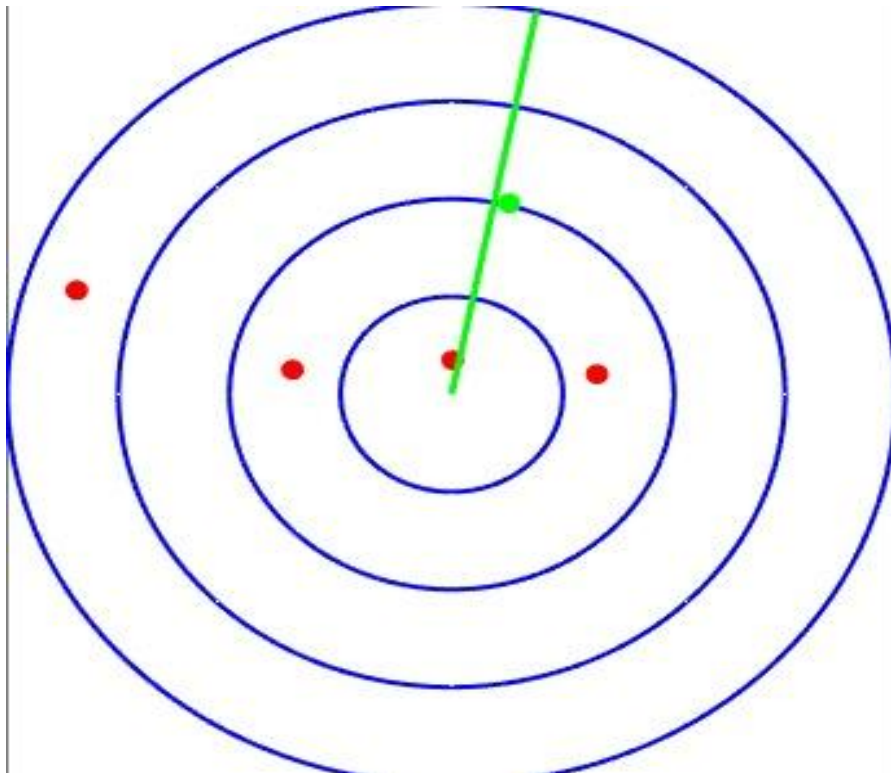
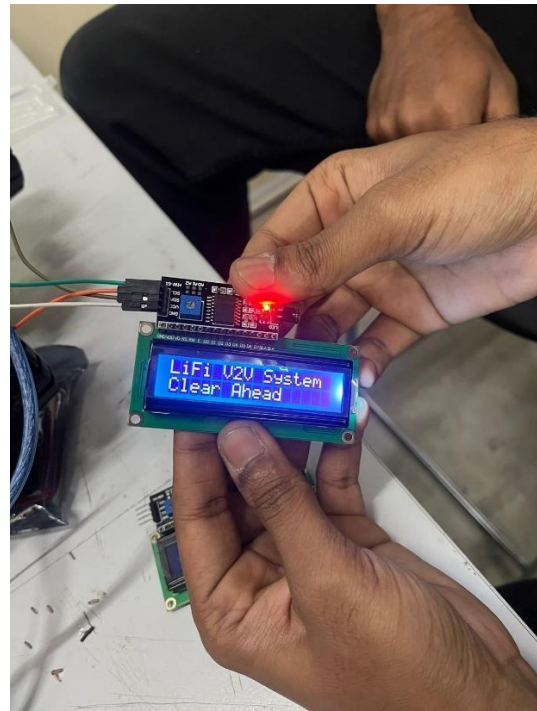
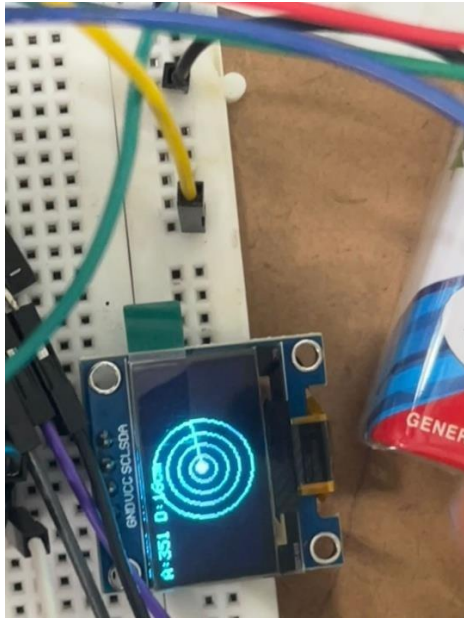
The experimental results obtained from the implementation of the Li-Fi Enabled Radar Based V2V Collision Prevention System demonstrate the effectiveness of integrating radar sensing with Li-Fi based vehicle-to-vehicle communication for enhancing road safety. The system performance observed during testing confirms that cooperative communication significantly improves collision prevention capability compared to standalone sensing systems.

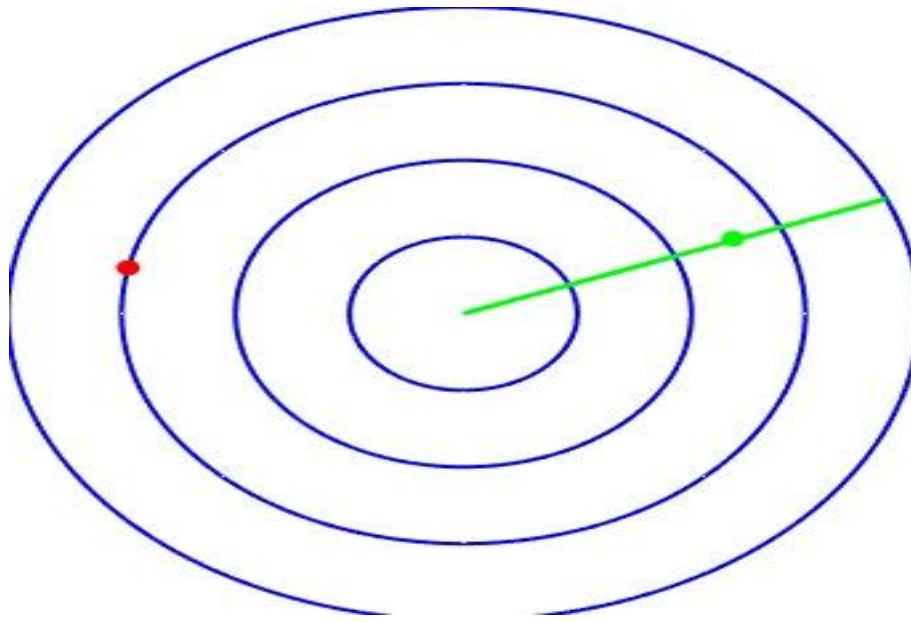
Radar sensing proved to be a reliable method for real-time obstacle detection and distance measurement. The consistency of radar readings across multiple test scenarios indicates that radar technology is well suited for collision detection under varying environmental conditions. However, radar alone relies on proximity thresholds, which may result in delayed response during sudden braking events. This limitation was effectively addressed by the inclusion of Li-Fi based V2V communication.

The Li-Fi communication module played a crucial role in reducing overall system response time. By enabling early transmission of braking and hazard information from the leading vehicle, the following vehicle received advance warning before a critical distance threshold was reached. This proactive information exchange enhanced situational awareness and allowed the braking mechanism to activate earlier, thereby reducing collision risk.

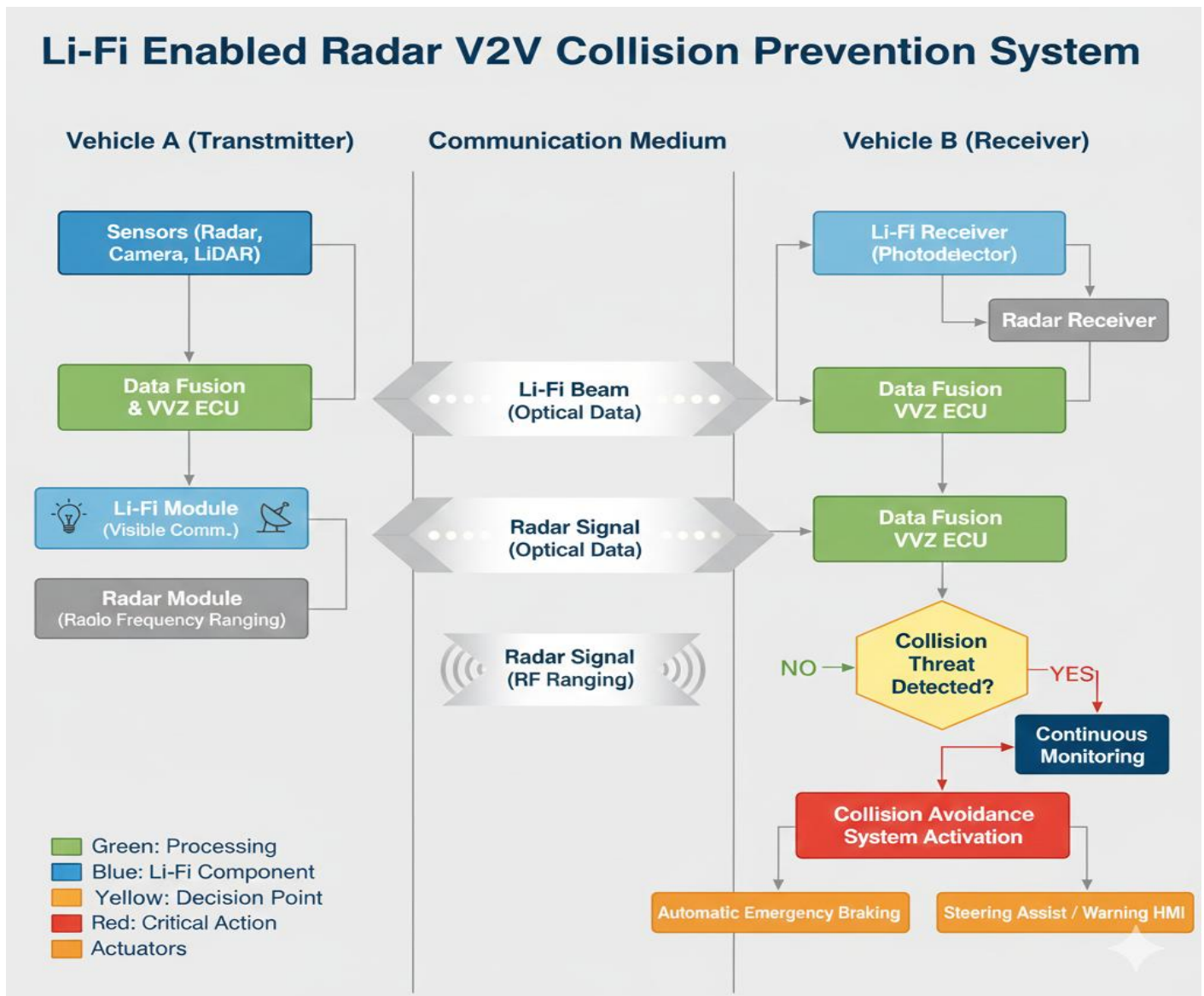
The integration of sensing and communication through a centralized control unit resulted in smooth and coordinated system behavior. The embedded control logic successfully balanced responsiveness and stability, minimizing false triggering while ensuring timely braking. The system demonstrated improved reliability by combining local sensor data with cooperative vehicle information, creating a multi-layer safety approach.

Overall, the discussion of results highlights that the proposed system offers a significant improvement in collision prevention effectiveness. The results validate the feasibility of using Li-Fi for V2V communication and confirm that combining it with radar sensing enhances









Chapter 6

APPLICATIONS AND ADVANTAGES

This chapter discusses the practical applications and key advantages of the Li-Fi Enabled Radar Based V2V Collision Prevention System. The proposed system is designed to enhance road safety by combining radar-based sensing with high-speed Li-Fi based vehicle-to-vehicle communication. Its applicability spans across modern transportation systems, intelligent vehicles, and smart city infrastructure.

6.1 Potential Applications

The proposed system has significant application potential in Intelligent Transportation Systems (ITS), where real-time data exchange between vehicles plays a crucial role in reducing traffic accidents and improving traffic flow. By enabling vehicles to communicate braking status and proximity information, the system supports cooperative safety mechanisms essential for modern ITS environments.

In smart vehicles, the Li-Fi enabled radar-based collision prevention system enhances existing safety features by providing proactive collision warnings and automated braking support. The integration of V2V communication allows smart vehicles to anticipate hazards beyond the line of sight, improving driver assistance and reducing dependence on human reaction time.

The system is also highly relevant for autonomous vehicles, where reliable sensing and communication are critical for safe operation. Autonomous vehicles rely on accurate environmental perception and inter-vehicle coordination. The proposed system provides an additional layer of safety by enabling direct and secure communication between autonomous vehicles, supporting cooperative decision-making and collision avoidance.

In the context of smart city traffic management, the system can be integrated with intelligent traffic infrastructure to improve overall road safety. Vehicles equipped with Li-Fi enabled V2V systems can interact with traffic signals, road-side units, and other vehicles to reduce congestion, prevent accidents at intersections, and support efficient traffic control strategies.

6.1.1 Intelligent Transportation Systems (ITS)

The proposed system plays a vital role in Intelligent Transportation Systems by enabling real-time communication between vehicles. By sharing safety-critical information such as braking

status, distance, and speed, the system helps reduce accidents, improve traffic flow, and support cooperative driving strategies. Integration with ITS infrastructure enhances overall transportation efficiency and safety.

Smart Vehicles

In smart vehicles, the system enhances existing safety features by providing advanced collision prevention capabilities. Radar-based sensing ensures accurate obstacle detection, while Li-Fi based V2V communication allows vehicles to anticipate hazards earlier. This improves driver assistance systems and reduces dependence on human reaction time.

Autonomous Vehicles

Autonomous vehicles require precise sensing and reliable communication for safe operation. The proposed system supports autonomous driving by providing an additional safety layer through cooperative V2V communication. Li-Fi enables secure and low-latency data exchange, which is essential for synchronized decision-making among autonomous vehicles.

6.2 Advantages of the Proposed System

One of the primary advantages of the proposed system is its ability to provide real-time vehicle-to-vehicle communication, allowing vehicles to exchange safety-critical information such as braking status and distance instantly. This cooperative communication significantly enhances situational awareness and enables faster response to potential hazards.

The use of Li-Fi technology offers low-latency, high-speed data transmission compared to traditional radio-frequency communication systems. Since Li-Fi operates using visible light, it is immune to electromagnetic interference and provides enhanced data security, making it suitable for safety-critical automotive applications.

The integration of radar sensing with Li-Fi based communication leads to improved collision prevention by combining accurate obstacle detection with early hazard notification. This dual-layer safety approach allows vehicles to react proactively, reducing the likelihood and severity of collisions.

Overall, the system contributes to enhanced road safety by reducing driver reaction time, minimizing human error, and enabling cooperative collision prevention. The cost-effective and scalable nature of the proposed system makes it suitable for widespread adoption in future connected and intelligent transportation systems.

6.2.1 Limitations and Comparative Perspective

While the Li-Fi Enabled Radar Based V2V Collision Prevention System demonstrates effective collision prevention through cooperative sensing and communication, certain limitations are inherent to the proposed approach. One of the primary limitations is the dependency of Li-Fi communication on clear line-of-sight conditions. Since Li-Fi uses visible light for data transmission, obstructions such as large vehicles, sharp curves, or adverse weather conditions can reduce communication reliability. This constraint limits its effectiveness in complex traffic scenarios where uninterrupted visibility cannot always be guaranteed.

Another limitation arises from the restricted communication range of Li-Fi compared to conventional radio-frequency based V2V systems. Although Li-Fi offers low latency and high data rates, its operational range is comparatively shorter, which may affect long-distance vehicle communication on highways. Additionally, precise alignment between Li-Fi transmitters and receivers is required for efficient data exchange, increasing system complexity.

From a sensing perspective, radar sensors, while reliable for distance measurement, may face challenges in distinguishing between multiple closely spaced objects or in environments with high reflection and noise. The accuracy of low-cost radar modules is limited compared to advanced automotive-grade sensors, which can impact system performance under complex traffic conditions. Furthermore, the prototype implementation relies on simplified braking mechanisms that may not fully replicate real-world vehicle braking dynamics.

From a comparative perspective, traditional collision prevention systems based solely on radar or camera sensors operate independently and lack cooperative awareness of surrounding vehicles. RF-based V2V systems, such as those using Dedicated Short-Range Communication (DSRC) or cellular technologies, offer longer range but are susceptible to interference, congestion, and security concerns. In contrast, the proposed system provides faster and more secure communication through Li-Fi, along with accurate radar-based detection, enabling proactive and cooperative collision prevention.

Although the proposed system has certain limitations, its advantages in terms of low latency, enhanced security, and cooperative safety make it a promising alternative to existing approaches. By addressing these limitations through future enhancements such as hybrid communication models and sensor fusion, the proposed system can outperform conventional collision prevention technologies and contribute significantly to intelligent transportation systems.

Chapter 7

CONCLUSIONS AND SCOPE FOR FUTURE WORK

This chapter presents a comprehensive conclusion of the Li-Fi Enabled Radar Based V2V Collision Prevention System, summarizing the objectives achieved, methodologies adopted, and results obtained throughout the course of the project. It evaluates the effectiveness of integrating radar-based sensing with Li-Fi enabled vehicle-to-vehicle communication in enhancing collision prevention and overall road safety. The chapter also discusses potential future enhancements aimed at improving the system's functionality, scalability, and adaptability to real-world traffic environments. Emphasis is placed on exploring advanced communication techniques, intelligent decision-making approaches, and infrastructure integration to extend the applicability of the proposed system toward next-generation intelligent and autonomous transportation systems.

7.1 Comparative Analysis

A comparative analysis is carried out to evaluate the effectiveness of the Li-Fi Enabled Radar Based V2V Collision Prevention System in comparison with conventional collision prevention approaches. Traditional vehicle safety systems primarily rely on standalone sensing technologies such as ultrasonic sensors, cameras, or radar to detect obstacles and potential collision threats. While these systems provide valuable assistance, they operate independently and lack cooperative awareness of surrounding vehicles, which can result in delayed response during sudden braking or unexpected maneuvers.

Camera-based systems offer visual information and object classification capabilities but are highly dependent on lighting conditions and are significantly affected by poor visibility, fog, or nighttime driving. Ultrasonic sensors are limited to short-range detection and are unsuitable for high-speed vehicular applications. Radar-based systems, although reliable in various weather conditions, depend solely on proximity thresholds and may not provide early warning of hazards originating from other vehicles.

Existing V2V communication systems based on radio-frequency technologies such as Dedicated Short-Range Communication (DSRC) or cellular networks enable cooperative safety mechanisms. However, these systems are prone to bandwidth congestion, electromagnetic

interference, latency issues, and potential security vulnerabilities, particularly in dense traffic environments.

In contrast, the proposed system combines radar-based sensing with Li-Fi enabled V2V communication, offering a dual-layer safety approach. Radar provides accurate and continuous distance measurement, while Li-Fi enables low-latency, high-speed, and secure data exchange using visible light. This integration allows vehicles to receive early braking and hazard information, significantly reducing reaction time and improving collision prevention effectiveness.

Overall, the comparative analysis demonstrates that the proposed system overcomes several limitations of conventional collision prevention technologies by enabling cooperative, low-latency, and interference-free communication. Although Li-Fi has certain limitations related to line-of-sight requirements, its integration with radar sensing provides a balanced and effective solution, making the proposed system a promising alternative for future intelligent transportation and connected vehicle applications.

7.2 Conclusions

The Li-Fi Enabled Radar Based V2V Collision Prevention System has been successfully designed and implemented to demonstrate an effective approach to cooperative vehicle safety. The proposed system integrates radar-based obstacle detection with Li-Fi enabled vehicle-to-vehicle communication, enabling real-time exchange of safety-critical information between vehicles. This integration addresses key limitations of conventional standalone collision prevention systems by enhancing situational awareness and reducing response time.

The experimental results confirm that radar sensing provides accurate and reliable distance measurement, while Li-Fi communication enables low-latency and secure data transmission. The embedded control unit effectively processes both local sensor data and received V2V information to identify potential collision scenarios and initiate timely braking actions. The coordinated operation of sensing, communication, and actuation components demonstrates the feasibility and reliability of the proposed system.

Overall, the project validates the effectiveness of combining radar sensing with optical wireless communication for collision prevention. The system offers a cost-effective, scalable, and

practical solution that can significantly enhance road safety by reducing collision risk, minimizing human error, and supporting intelligent transportation systems. The successful implementation of the prototype confirms the potential of the proposed approach for future connected and smart vehicle applications.

7.2.1 Scope for Future Work

Several The Li-Fi Enabled Radar Based V2V Collision Prevention System presented in this project establishes a strong foundation for cooperative vehicle safety; however, several opportunities exist for further enhancement and extension. One of the primary areas for future work is the development of multi-vehicle communication frameworks, where multiple vehicles can simultaneously exchange safety-critical information. Such an approach would enable coordinated collision prevention in dense traffic conditions and improve overall traffic flow management.

Future work can also focus on integrating artificial intelligence and machine learning algorithms into the decision-making process. These techniques can be used to analyze driving patterns, predict potential collision scenarios, and dynamically adjust safety thresholds based on traffic density, vehicle speed, and environmental conditions. This would significantly improve system adaptability and reduce false alerts.

Another important direction for future research is the implementation of advanced autonomous braking systems. By integrating the proposed system with automotive-grade braking control units, fully automated emergency braking can be achieved, reducing dependence on driver intervention and improving safety in high-speed and critical scenarios.

The system can further be enhanced by enabling vehicle-to-infrastructure (V2I) communication, allowing interaction with smart traffic signals, road-side units, and intelligent traffic management systems. This integration would support smart city initiatives, improve congestion control, and enhance situational awareness beyond vehicle-to-vehicle communication.

Additionally, future work may explore hybrid communication models that combine Li-Fi with radio-frequency technologies to overcome line-of-sight limitations and ensure reliable communication under all traffic conditions. Improvements in sensor fusion, power management, and hardware miniaturization can further increase system efficiency and suitability for real-world deployment.

Overall, these future enhancements can transform the proposed system into a comprehensive intelligent transportation solution capable of supporting next-generation connected and autonomous vehicles.

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APPENDIX

- A.1 Complete Embedded Code for Li-Fi Enabled Radar System
 - Sensor initialization for Radar/Ultrasonic units.
 - Li-Fi communication protocols and data exchange logic.
 - Control logic for Warning alerts and Automatic Braking.
- A.2 Circuit Diagram and Hardware Layout
 - Complete circuit schematic for the Arduino/Microcontroller.
 - Interfacing between Radar sensors, Li-Fi transmitter/receiver modules, and braking actuators.
- A.3 Extended Test Results and Tables
 - Tabulated distance measurements and communication latency results.
 - Detailed packet delivery ratios and data throughput versus distance.